

Kaden Alvarino

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Professional Summary

Hands-on Aerospace Engineering graduate with practical experience spanning aircraft design and fabrication, CAD modeling, and embedded systems. Designed and built a 25 lb VTOL drone, developed lithography and microfabrication processes in semiconductor and quantum computing research, and worked on autonomous flight systems for designated missions. Core interests in aerodynamics, flight controls, systems, and fabrication. Fluent in Spanish; willing to relocate.

Education

- B.S. in Aerospace Engineering** May 2026
North Carolina State University · Raleigh, NC
- M.S. in Aerospace Engineering** May 2028
Virginia Polytechnic Institute · Blacksburg, VA

Relevant Work Experience

- Mechanical and Design Engineering Intern** May 2026 – Aug 2026
Legrand · Concord, NC
 - Designed and fabricated machining solutions to improve reliability and operator satisfaction.
 - Modeled existing equipment in SolidWorks to implement new technology with usable hardware.
 - Collected user feedback to improve safety and efficiency in different departments across the plant.
- Process Engineering Intern** May 2025 – Aug 2025
Qorvo · Greensboro, NC
 - Calibrated photoresist spinner equipment and computed spin curves to improve thickness consistency.
 - Developed and optimized lithography processes to maximize wafer yield while increasing production throughput.
 - Served as Intern Engagement Leader, organizing outings and networking events for 54 interns across the summer.
- Undergraduate Researcher in Physics** May 2024 – Jul 2024
University of Arkansas · Fayetteville, AR
 - Conducted superconductor research toward quantum computing device fabrication.
 - Led development of an air-bridge structure to induce charge through air as a medium in a quantum refrigerator.

Other Experience

- DARPA Lift Challenge — Lead Engineer** Jan 2026 – May 2026
North Carolina State University · Raleigh, NC
 - Collaborated with student engineers to design an aircraft capable of carrying a payload 4× its own weight.
 - Applied undergraduate knowledge of construction and structural testing to validate design prior to build.
- Senior Design Capstone Project** 2025 – 2026
North Carolina State University · Raleigh, NC
 - Designing and constructing a 10ft 25lb VTOL aircraft on a \$1,500 budget for the NC Forestry Service.
 - Produced documentation covering mission intent, design rationale, risk evaluation, and feasibility evidence.
- College Mentors for Kids** Ongoing
North Carolina State University · Raleigh, NC
 - Mentored elementary school students weekly on campus, supporting reading, writing, and artistic development.

Skills

- CAD** SolidWorks, OpenVSP, AutoCAD, Fusion 360, KLayout
- Software** MATLAB, Microsoft Excel, ANSYS (beginner)
- Languages** Spanish (Fluent), Italian (Beginner)